MIL-C-23136B
9 June 1977
SUPERSEDING
MIL-C-23136A
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MILITARY SPECIFICATION

CONDENSERS, REFRIGERATING, WATER-COOLED, REFRIGERANT-12

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers water-cooled refrigerating condensers.

1.2 Classification. Water-cooled condensers shall be of the following types and sizes, as specified (see 6.2):

Type I	-	Closed shell-and-tube
Type II	-	Closed shell-and-coil
Type III	-	Double tube

		Heat Rejection
Size 1	-	25,000 Btu per hour (7 kw)
Size 2	-	37,500 Btu per hour (11 kw)
Size 3	-	50,000 Btu per hour (15 kw)
Size 4	-	65,000 Btu per hour (19 kw)
Size 5	-	85,000 Btu per hour (25 kw)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4130

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2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-C-650 - Crates, Wood, Open and Covered.
PPP-T-60 - Tape: Packaging, Waterproof.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-760 - Condensers, Refrigerating, Water-Cooled, Refrigerant-12.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

Standard No. 22 - Methods of Rating and Testing Water-Cooled Refrigerant Condensers.

(Application for copies should be addressed to the American Society for Heating, Refrigerating and Air-Conditioning Engineers, 345 East 47th Street, New York, NY 10017.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N. W., Washington, DC 20036.)

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UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

UNDERWRITERS LABORATORIES INCORPORATED (UL)

UL 207 - Refrigerant Containing Components and Accessories, Nonelectrical

(Application for copies should be addressed to the Underwriters Laboratories Inc., 207 East Ohio Street, Chicago, IL 60611; 333 Pfingston Rd., Northbrook, IL 60062; Santa Clara, CA 95050.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish a sample of the item to be procured under this specification for first article inspection and approval (see 4.3 and 6.3).

3.2 Standards.

3.2.1 UL standard. The water-cooled condensers shall be constructed according to the UL Standard No. 207.

* 3.2.1.1 UL certification. Prior to commencing production, the contractor shall submit to the contracting officer or his authorized representative satisfactory evidence that the water-cooled condensers meet the requirements of UL Standard No. 207. Acceptable evidence of meeting this standard shall be a UL listing mark or label, or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the condensers offered have been tested and conform to UL Standard No. 207.

* 3.3 Materials. Materials not definitely specified shall be of the quality normally used by the manufacturer for his water-cooled condensers provided the completed item complies with all provisions of this specification. Unless otherwise specified herein, commercial tolerance shall apply to metal thicknesses and diameters.

3.4 Design and construction. The design and construction of the water-cooled condensers covered by this specification shall conform to the requirements herein and the design, details, and dimensions thereof shown in MIL-STD-760. The condensers shall conform to the applicable capacity pressure drop and water-flow rate requirements of MIL-STD-760, when tested as specified in 4.5.1.

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3.5 Finish. The water-cooled condensers shall have the manufacturer's standard commercial finish.

3.6 Marking. Water-cooled condensers shall be marked as specified in MIL-STD-760 and in accordance with UL Standard No 207.

* 3.7 Workmanship. The finished water-cooled condensers shall be clean and all forming and cutting shall be done accurately. All scale and flux deposits shall be removed from the finished welds. All exposed welds shall be ground smooth to match the adjacent surfaces. All nuts, bolts, and srews shall be tight. There shall be no rough or sharp edges, slivers, or burrs on the finished water-cooled condensers.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3)
2. Quality conformance inspection (see 4.4)

* 4.3 First article. When a first article is required, the water-cooled condensers shall be examined for defects listed in table I, for dimensions specified and tested as specified in 4.5.1 Failure to pass the test, any defects any dimension not as specified shall constitute failure of the first article.

4.4 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

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4.4.2 End item inspection. The lot shall consist of all condensers of one type and size offered for inspection at one time. The sample unit shall be a completely assembled condenser.

4.4.2.1 Visual examination. Examination shall be made of the condenser for the defects listed in table I. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not manufacturer's standard finish		X
Construction and workmanship	Part missing, or not specified type or size	X	
	Fractured, split, punctured, sprung, malformed, or otherwise impaired	X	
	Sharp burr or sliver on surfaces to be handled during operation	X	
	Not connected or joined as specified	X	
Welding	Missing, incomplete, not smooth and uniform, scale or flux deposit not removed	X	X
Bolts, nuts, screws,	Missing, broken, stripped or loose	X	
Marking	Missing, incomplete, not legible		X

4.4.2.2 Dimensional examinations. Examination shall be made for dimensions specified. Any deviation from specified requirements shall be considered a defect. The inspection level shall be S-2 with an AQL of 6.5, expressed in terms of defects per hundred units.

4.4.2.3 End item testing. The first article or, if none is required, the initial production unit shall be tested as specified in 4.5.1. Any nonconformance shall be cause for rejection of the condensers.

* 4.4.3 Packaging inspection. An examination shall be made to determine that packaging, packing, and markings complies with section 5 requirements. Defects shall be as specified in table II. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be closed. The lot shall be the number of containers offered for delivery at one time. The inspection level shall be S-2 and the AQL shall be 4.0, expressed in terms of defects per hundred units.

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TABLE II. Examination of packaging

Examination	Defect
Marking	Missing, incorrect, illegible, of improper size, location, sequence, or method of application.
Materials	Component missing, damaged, or otherwise defective.

4.4.3.1 Examination of shipping containers. When shipping containers are required to be in accordance with PPP-C-650 or PPP-B-601, examination for defects in anchoring shall be in accordance with PPP-C-650 and examination for defects in closure and strapping shall be in accordance with the appendix of the applicable container specification.

* 4.4.4 Certification examination. Certificates of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with the specification requirements shall be examined and validated.

4.5 Test method.

4.5.1 Capacity, pressure drop, and water-flow rate. Water-cooled condensers shall be tested for capacity, for pressure drop, and for water-flow rate to determine compliance with MIL-STD-760 (see 3.4) using test methods described in ASHRAE Standard No. 22.

* 5. PACKAGING

* 5.1 Preservation-packaging. Preservation-packaging shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. All openings into the condenser shall be sealed with tape conforming to type IV, class 1 of PPP-T-60 or commercially used plugs or caps.

5.1.2 Commercial. The condenser shall be packaged to afford adequate protection against damage during shipment from the contractor to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

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5.2.1 Level A packing. Each condenser, preserved-packaged as specified in 5.1, shall be packed in a shipping container conforming to type II, style A of PPP-C-650 or overseas type of PPP-B-601. Anchoring of contents shall be in accordance with the appendix of PPP-C-650. Closure and strapping shall be in accordance with the appendix of the applicable container specification.

5.2.2 Level B packing. Each condenser, preserved-packaged as specified in 5.1, shall be packed in a shipping container conforming to type II, style B of PPP-C-650 or domestic type, style A or B of PPP-B-601. Anchoring of contents shall be in accordance with the appendix of PPP-C-650.

* 5.2.3 Commercial packing. The condenser, preserved-packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall comply with the Uniform Freight Classification or National Motor Freight Classification, as applicable.

* 5.3 Marking. In addition to any special marking required by the contract or order, shipping container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The water-cooled condensers covered by this specification are intended for use with Military standard open type and hermetic compressors in refrigeration and air conditioning systems.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and size of water-cooled condensers required (see 1.2).
(c) When a first article is required (see 3.1, 4.3 and 6.3).
(d) Selection of applicable levels of preservation-packaging and packing (see 5.1 and 5.2).

* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample, initial production item or other specific item described under the definition of a First Article in the ASPR. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

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6.4 Changes from previous issue. The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - YD
Air Force - 99

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OMB Approval
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MIL-C-23136B - Condensers, Refrigerating, Water-Cooled, Refrigerant-12

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

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